

Exploratory Data Analysis

Discover Stories Hidden in Your Data

■ EDA MASTERY

■ LEARNING OUTCOME

By the end of this PDF, you will conduct a full EDA: summarize data, find patterns, detect outliers, and generate hypotheses using Python visualization libraries.

■ Skills You Will Gain

- Follow a structured EDA workflow
- Use `.describe()`, `.info()`, `value_counts()`
- Create histograms, box plots, scatter plots
- Detect outliers using IQR and Z-score
- Analyze correlations with heatmaps
- Generate and document data insights

■ EDA is where real insights are born. Companies pay analysts specifically to 'explore and find what we don't know we're missing'.

SECTION 1

EDA Workflow (7 Steps)

1	Load & Inspect pd.read_csv() → df.head(), df.info(), df.shape
2	Check Quality Missing values, duplicates, wrong types
3	Univariate Analysis Study each column individually: distributions, outliers
4	Bivariate Analysis Relationships between pairs of variables
5	Multivariate Analysis Correlation matrix, pair plots, 3D scatter
6	Outlier Detection Box plots, IQR method, Z-score
7	Document Insights Write down every observation with supporting chart

SECTION 2

First Look at Your Data

```
import pandas as pd import matplotlib.pyplot as plt import seaborn as sns df =  
pd.read_csv('sales.csv') df.head() # First 5 rows df.tail() # Last 5 rows df.shape # (rows,  
columns) df.info() # Types + null counts df.describe() # Stats: count, mean, std, min, Q1,  
Q2, Q3, max df.nunique() # Unique values per column df['Category'].value_counts() # Frequency  
of each category
```

SECTION 3

Univariate Analysis

For Numeric Columns (histogram + box plot):

```
# Histogram – see distribution shape plt.figure(figsize=(10, 4)) plt.subplot(1,2,1)  
df['Revenue'].hist(bins=30, color='steelblue') plt.title('Revenue Distribution') # Box Plot –  
see outliers and quartiles plt.subplot(1,2,2) df.boxplot(column='Revenue') plt.title('Revenue  
Box Plot') plt.show()
```

For Categorical Columns (bar chart):

```
df['Region'].value_counts().plot(kind='bar', color='coral') plt.title('Orders by Region')
plt.xlabel('Region') plt.ylabel('Count') plt.show()
```

SECTION 4

Bivariate Analysis

```
# Scatter plot – numeric vs numeric plt.scatter(df['StudyHours'], df['Score'], alpha=0.5)
plt.xlabel('Study Hours') plt.ylabel('Exam Score') # Box plot by category – numeric vs
categorical sns.boxplot(x='Region', y='Revenue', data=df) # Grouped bar chart
df.groupby('Category')['Revenue'].mean().plot(kind='bar')
```

SECTION 5

Correlation Heatmap

```
plt.figure(figsize=(10, 8)) corr_matrix = df.select_dtypes(include='number').corr()
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', center=0, vmin=-1, vmax=1)
plt.title('Correlation Heatmap') plt.tight_layout() plt.show()
```

SECTION 6

Outlier Detection

```
# IQR Method Q1 = df['Salary'].quantile(0.25) Q3 = df['Salary'].quantile(0.75) IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR upper = Q3 + 1.5 * IQR outliers = df[(df['Salary'] < lower) |
(df['Salary'] > upper)] print(f'Found {len(outliers)} outliers') # Z-Score Method from scipy
import stats z_scores = stats.zscore(df['Salary']) outliers_z = df[abs(z_scores) > 3]
```

■ PRACTICE QUESTIONS

Q1. What does `df.describe()` show?

Answer: Count, mean, std, min, Q1 (25%), median (50%), Q3 (75%), and max for all numeric columns.

Q2. What does a right-skewed histogram tell you?

Answer: Most values are low, but a few very high values pull the tail to the right. Mean > Median.

Q3. You find correlation = 0.82 between age and income. What does this mean?

Answer: Strong positive correlation — older people tend to have higher income in this dataset.

Q4. How do you detect outliers using the IQR method?

*Answer: Calculate Q1, Q3, IQR = Q3-Q1. Outliers are values below Q1-1.5*IQR or above Q3+1.5*IQR.*

Q5. What chart would you use to compare salary distribution across 5 departments?

Answer: Box plot — it shows median, Q1, Q3, and outliers for each department side by side.

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: customers.csv (Age, Gender, Income, SpendingScore, Region, Tenure)

Steps:

1	Load Read CSV, run .info(), .describe(), check nulls
2	Univariate Histogram of Age, Income; bar chart of Region and Gender
3	Bivariate Scatter: Income vs SpendingScore; box plot: SpendingScore by Region
4	Correlation Heatmap of all numeric variables
5	Outliers Detect outliers in Income using IQR method
6	Insights Write 5 observations from your analysis

Expected Output: EDA report: 6+ charts + 5 written insights in Python notebook

Insights to Find: Who spends the most? Any correlation between income and spending? Any outlier customers?

EDA Python Cheat Sheet	
<code>df.head() / df.tail()</code>	Preview first/last rows
<code>df.info()</code>	Types, nulls, memory usage
<code>df.describe()</code>	Summary statistics
<code>df.nunique()</code>	Unique values per column
<code>df['col'].value_counts()</code>	Frequency of each category
<code>df.isnull().sum()</code>	Missing values count
<code>df.corr()</code>	Correlation matrix
<code>sns.heatmap(df.corr())</code>	Visualize correlations
<code>df['col'].hist()</code>	Distribution histogram
<code>sns.boxplot(x, y, data)</code>	Box plot by category
<code>plt.scatter(x, y)</code>	Scatter plot
<code>df.groupby('col').agg()</code>	Group & aggregate